DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
Faculty of Science & Technology
Bangladesh University of Professionals (BUP)
Course Title: **Digital Communication (ICE 3105)** (3rd Year, 1st Semester, January-June, 2026)
Class Test-03

Time: **20 minutes** Full Mark: **20**

1. A Digital Communication system uses the CRC method to encode and decode its messages. Given the following data word and generator polynomial, find the code word at the transmitter part. 5
   - Data Word: 1011
   - Generator polynomial, $G(x) = x^3 + x^2 + 1$

2. In the Hamming Code, A Syndrome of a particular number of bits, S, and an extra parity bit P_0 is used to detect and correct errors. Explain the types of errors that the combinations of S and P_0 can provide us. 5

3. A Digital Communication system uses the LDPC (6,3) method to encode and decode its messages. Find out the correct data word from the code word received at the receiver end. Note that there is more than one error in the code. 10
   - Code Word: 1 1 0 0 0 0
   - Parity Check Matrix $H(x) = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$